

Education

Singapore University of Technology and Design, Bachelor of Engineering in Computer Science, 2026.

- **Cocurriculars:** Galois Group Mathematics Society, Google Developer Student Club (3DC).
- **Concentrations:** Minor in Artificial Intelligence, Specialisation in Visual Analytics and Computing.

Experiences

Creatz3D and Figure Factories, AI Engineer, 2025.

- Accelerated ML training pipelines by changing primitives and vectorising functions.
- Operationalised mesh generation by creating a cron-scheduled Hunyuan3D-2 inference pipeline.
- Productionised mesh thickening by training, testing, and deploying GraphSAGE models to thicken meshes.
- Supported ML inference pipelines by patching CUDA kernels for dependency compatibility.
- Streamlined UGC production by designing human-in-the-loop generative AI orchestrations.

Proficiencies

- **Languages.** C, CSS, Dockerfile, HTML, Java, JavaScript, Makefile, Python, Shell, SnekScript, Swift, TeX.
- **Tools.** AWS, Azure, Cloudflare, Docker, FAISS, FastAPI, Figma, Firebase, Flask, Git(Hub), Gradio, Hugging Face, LangChain, llama.cpp, Matplotlib, MLX, Next.js, NumPy, Ollama, OpenRouter, Open WebUI, Paint.NET, PyTorch, Qdrant, React, SQLite, Shadcn, Streamlit, Tailwind, TensorFlow, Vercel, Whisper, n8n, pandas, scikit-learn.

Projects

PDPal, AI Legal Assistant, 2026.

- Built and refined backend architecture by running experiments and reasoning over trade-offs.
- Validated system performance with oracle@3 beating GPT-5.4 by 4.5% and Gemini 3.1 Pro by 9.1% on YPDYC.
- Improved frontend accessibility by designing responsive multilingual visual components.
- Validated frontend accessibility with 73% of users fully recommending the product in preliminary studies.

SUTD404, SUTD Room Address Lookup Tool, 2026.

- Trained and tested Chinchilla-optimal Nanochat-style transformers to rewrite noisy SUTD room names.
- Validated model performance with accuracy outperforming SUTD Assistive Maps on a test subset.
- Optimised perceived and actual speed of inference by making it cached, debounced, and incremental.

Snekboros, "Vision World Models for Snake Playing", 2026.

- Trained and tested SIGReg-ularised convolutional GRU world models to play 1P Snake from 4x4 grid images.
- Demonstrated the effect of planning horizon and sample count on survival.

Vibe Decoding, "An Unexpected Expected Next-Token Rule for Language Models", 2025.

- Proposed aligning next-token embeddings with their expectation.
- Achieved a 7.7% macro F1 increase on BoolQ with Qwen 2.5 0.5B Instruct.

BLIPS Regularisation, "Regularisation with Buffered Linear Interpolation by Positional Similarity", 2025.

- Proposed using a buffer to linearly interpolate between recent and current features by positional similarity.
- Trained and tested convolutional neural networks on a STEAD subset.
- Achieved a 94.5% macro F1 on a STEAD subset with a convolutional vision transformer.

Recognition

- Showcased MLOps Projects, Singapore University of Technology and Design, 2026.
- Showcased Deep Learning Projects, Singapore University of Technology and Design, 2025.
- Information Systems Prize, Singapore University of Technology and Design & Singtel, 2024.
- **Physical World Design 1D Award**, Singapore University of Technology and Design, 2022.
- Malay (Special Programme) Top in Cohort Award, Temasek Secondary School, 2018.
- Mathematics and Science Book Prize, Temasek Secondary School, 2017.